

JAYPEE INSTITUTE OF INFORMATION TECHNOLOGY

PROJECT SYNOPSIS
ON
EXPENSE TRACKER SYSTEM
USING C++

INTRODUCTION TO OPEN-SOURCE DEVELOPMENT

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ABOUT THE PROJECT

The Expense Tracker System is a console-based application developed using C++ that helps users manage and monitor their daily expenses efficiently. The system enables users to record, update, categorize, and track their expenses while maintaining a monthly budget. It also provides reports and analytics to help users understand their spending patterns and make informed financial decisions.

The project will be developed with open-source software development practices using Git and GitHub. The development process will include repository management, branching strategies, pull requests, code reviews, issue tracking, documentation, and version management. This approach will provide practical exposure to collaborative software development.

The application will be designed using Object-Oriented Programming (OOP) principles, Standard Template Library (STL), and file handling techniques.

OBJECTIVES

- Develop a user-friendly Expense Tracker application using C++.
- Implement OOP concepts such as classes, objects, encapsulation, abstraction, and modular design.
- Provide functionality for adding, updating, deleting, and viewing expenses.
- Categorize expenses into groups such as Food, Travel, Education, and Entertainment.
- Enable users to set monthly budgets and monitor spending.
- Generate reports and spending summaries.
- Perform expense analytics such as highest expense, lowest expense, and category-wise spending.
- Store and retrieve data using file handling techniques.
- Gain practical experience with Git and GitHub workflows.
- Understand collaborative open-source software development practices.

REQUIREMENTS

- **Software Requirements:** C++, Visual Studio Code/Code::Blocks/Dev-C++, Git, GitHub, Windows.
- **Functional Requirements:** User Registration/Login, Expense Management, Categorization, Budget Management, Reports, Analytics, Search & Filter, File Storage.
- **Non-Functional Requirements:** Modular Code, Documentation, Version Control Compliance, Collaboration Support, Reliable Data Storage.

EXPECTED OUTCOME

The proposed system will provide users with an efficient way to manage personal finances by recording expenses, tracking budgets, and generating useful reports. The project will demonstrate practical implementation of C++ programming concepts, file handling, software design principles, and open-source development methodologies.